# M.AI.ESTRO- Twin Vision Pipeline Placement Report

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**Unit** ARI5907 Placement in AI
**Placement Title** M.AI.ESTRO-Twin: Automated Digital Twin Synthesis for Industrial Skill Transfer
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## Abstract

This report presents the final outcome of the ARI5907 placement, which contributed a vision perception pipeline to the M.AI.ESTRO-Twin project. The placement addressed the problem of generating structured perception signals automatically from egocentric manipulation demonstrations without manual prompt entry. Qwen3-VL-2B-Instruct paired with SigLIP was selected as the production configuration, achieving a semantic quality score of 0.365 at 1.432 seconds per window, the best latency adjusted result across all candidates. The motivating context was twofold: raw video based world model pipelines can encode task-irrelevant visual variation, and open-set perception stages normally require human-supplied object prompts before grounding can begin. Six vision language models were benchmarked on a shared 64-window sample drawn from the EgoDex dataset, with SigLIP and OpenCLIP compared as semantic quality backends. The semantic stage was extended into a downstream grounding path in which object lists from the semantic manifest became automated prompts for Grounding DINO and SAM 2, producing aligned bounding boxes and binary masks. The resulting pipeline yields structured manifests containing captions, tags, object prompts, embeddings, spatial detections, and segmentation masks that serve as a conditioning layer for subsequent world model and policy training stages. The 6 DoF stage was not carried forward because the semantic and 2D grounding stages had already reached a useful prototype state, whereas a geometry aware extension would have introduced substantially more complexity with limited immediate value. The primary contribution of this placement is a reusable, inspectable AI vision layer that advances the M.AI.ESTRO-Twin pipeline from raw egocentric video to machine readable task state representations.

## 1 Introduction

The placement was carried out alongside Research Support Officer duties linked to Project M.AI.ESTRO, an active Malta based research programme developing automated digital twin technology for industrial skill transfer. The declared placement title remained *M.AI.ESTRO-Twin: Automated Digital Twin Synthesis for Industrial Skill Transfer*, and the motivating problem was how to generate structured perception signals automatically from expert demonstrations without the need for manual prompt entry or annotation.

Two related problems motivated the study. The first was the broader representation problem faced by video and world model pipelines. These systems compress raw visual streams into latent state representations, but raw video includes task relevant dynamics alongside nuisance variation such as camera motion, lighting change, background motion, and object movements that are not conditioned on the target skill. This creates a familiar shortcut-learning risk in visual modelling: a system can rely on visually available but task-irrelevant correlations when those correlations are easier to encode than the intended causal structure [1]. In a dash camera setting, for example, a poorly conditioned representation may spend capacity modelling sky, tree, or camera-motion regularities rather than the traffic behaviour needed for driving. In the placement context, object lists, action descriptors, boxes, and masks were therefore treated as enrichment signals that help focus downstream learning on the manipulated objects and task relevant state changes.

The second problem was more implementation specific. Open-set perception tools can localise arbitrary text-conditioned objects, but they normally require a prompt or object list before localisation begins. In a manually operated workflow, that prompt would be supplied by a human annotator or user. Figure 1 illustrates this prompt-conditioned stage. The placement addressed this bottleneck by using the vision language model semantic stage to generate the object list automatically, so that the same sampled EgoDex window could drive semantic description, object prompting, grounding, and mask generation without manual prompt entry.

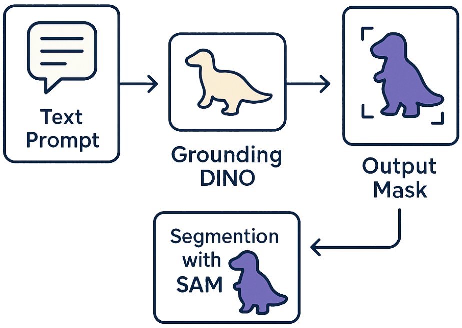


**Figure 1:** Prompt-conditioned grounding problem addressed by the placement. Open-set detectors can localise text-specified objects, but the text prompt normally has to be supplied before detection can begin.

The one page progress report submitted at the midpoint of the placement established that a working semantic alignment pipeline had been implemented and that an initial vision language model comparison had been completed. This final report continues from that checkpoint by documenting how the semantic stage was refined, how downstream grounding and mask stages were added, and why the placement delivered a 2D perception and conditioning layer rather than the full four stage pipeline originally proposed.

The aim of this placement was to develop a reusable AI vision pipeline that transforms raw egocentric manipulation data into machine readable representations of task relevant state, suitable for conditioning downstream learning modules within M.AI.ESTRO. To achieve this aim, five objectives were pursued:

1. Establish a semantic alignment pipeline capable of generating structured object and action descriptors from EgoDex manipulation windows.
2. Evaluate and select a vision language model and embedding backend appropriate for automated semantic quality assessment.
3. Extend the semantic stage into a downstream grounding pipeline using open-set object detection and segmentation.
4. Produce aligned structured manifests containing captions, tags, object prompts, bounding boxes, and masks suitable for world model conditioning.
5. Document the design decisions and model selection rationale in a form that supports future M.AI.ESTRO pipeline development.

The central research question guiding the implementation was: *Can vision language model object lists serve as reliable automated prompt sources for downstream open-set object grounding in egocentric manipulation data?* The original proposed stages were semantic understanding, bounding box grounding, binary mask generation, and 6 DoF pose estimation. As the implementation matured, the scope was deliberately narrowed to the first three stages because they formed the perception enrichment layer most directly needed by the downstream world model work. The report is structured as follows:

- Section 2 describes the methodology.
- Section 3 presents the results.
- Section 4 reports placement effort.
- Section 5 draws conclusions and identifies future work.

## 2 Methodology

All data used in this placement was drawn from EgoDex, a publicly released dataset; no human participants were recruited, no personal data were collected, and no ethical approval beyond standard academic data use was required. EgoDex is a large scale egocentric manipulation dataset comprising approximately 1.1 million demonstration clips across 194 object categories designed for dexterous interaction learning [2]. The placement did not process the full dataset. It used a local subset drawn from EgoDex training parts 1, 2, and 3; the fixed-stride index for this subset contained 430,962 candidate windows, from which the shared 64-window benchmark slice was sampled for the semantic comparison. The implemented pipeline operated on indexed windows extracted from that subset rather than processing every frame independently. Since the source videos run at 30 frames per second, frame-by-frame semantic inference would have introduced an impractical amount of redundant computation. A fixed windowing strategy shown in Figure 2 was therefore adopted as the first operational approach, in which each sample contained 16 observation frames and 16 prediction frames at a stride of 128 frames. At 30 fps and a 128-frame stride, consecutive windows are separated by approximately 4.3 seconds, a span sufficient for meaningful object state changes to emerge in manipulation tasks while keeping the index size tractable for placement stage work. As work within the broader M.AI.ESTRO research programme progressed, this fixed stride scheme was subsequently superseded by an adaptive segmentation strategy that divides each episode into a configurable number of relative position segments and applies a variable frame skip proportional to episode length, achieving substantially better coverage across variable duration demonstrations. The transition from the initial fixed stride approach to this adaptive extraction method therefore reflects a direct line of development that originated in the placement prototype.

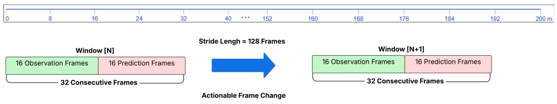


**Figure 2:** Fixed stride window sampling used for placement stage preprocessing. Each indexed sample contains 16 observation frames and 16 prediction frames.

### 2.1 Semantic Alignment

The first semantic candidate used during development was Qwen3-VL-8B-Instruct, selected because the early project goal required a model capable of generating scene level descriptions and structured semantic signals from a small set of representative frames [3]. The initial implementation was tightly coupled to that model, with prompt templates, output parsing, and quality checks written specifically for its response format. Once the semantic stage was shown to function reliably end-to-end, the code was refactored into a model agnostic benchmark contract with a shared prompt interface, standardised output schema, and interchangeable embedding backend. This refactoring was essential because the objective was not simply to make the pipeline run once, but to produce a reusable and comparatively inspectable semantic stage.

In this context, embeddings are the vector representations used to place text and visual content into a common comparison space [4]. The semantic generator produced captions and structured tags, while the embedding backend was used to assess how well those generated descriptions still corresponded to the sampled window frames. The embedding model was therefore not merely an implementation detail, but part of the semantic quality check loop itself. SigLIP became the main operational backend because its pairwise sigmoid training objective produces per-pair similarity scores that do not depend on batch composition, making it well suited to the role of an independent semantic quality check on individual window caption pairs [5]. The project later also explored OpenCLIP as an alternative similarity model in order to test whether the semantic ranking remained stable under a different comparison space [6].

The candidate generator set then widened. The Qwen family was compared with LLaVA-v1.6 [7], SmolVLM2-2.2B-Instruct [8], and Idefics3-8B [9]. In practical terms, the Qwen family showed the strongest overall balance for this task. During this benchmarking stage, the release of the Qwen 3.5 family provided an opportunity to extend the comparison further and examine whether the new variants changed the quality versus latency trade off enough to justify a change in the operational pipeline [10, 11]. These later checks became part of the placement process rather than a separate line of work.

The output schema of the semantic stage was refined iteratively as the pipeline moved from human readable inspection artefacts to machine parseable structured manifests. Initially, the stage emphasised raw artefacts so that the output could be inspected directly and used to judge whether the model was describing the task window sensibly. As the placement moved towards a multi stage pipeline, human readable artefacts alone were no longer sufficient: the grounding stage required a consistent, machine parseable object list rather than free text captions to function without manual intervention. The outputs therefore had to be written in a structured schema so that the next stage could consume them reliably. For this reason, the semantic stage was revised to write an aligned manifest, captions, tags, embeddings, and especially the object list information that acts as the prompt source for downstream grounding. The remaining manifest fields are also useful later as richer conditioning signals, particularly when contact relations or state changes are detected in the window.

### 2.2 Bounding Box Grounding

Once the semantic stage had stabilised, its outputs were used to drive the next stage of the pipeline. Rather than entering text prompts manually, prompt candidates were derived from the semantic results themselves, with particular attention to the contact object and the deduplicated object list. This design choice preserved the pipeline logic established in the semantic stage: the model first interpreted the task window and then supplied the vocabulary required for localisation.

Those prompts, exemplified in Figure 3, were passed into Grounding DINO, which produced text conditioned bounding boxes [12]. This step was important because the placement was not limited to scene description and had to move from semantic interpretation towards explicit image space localisation. The resulting boxes therefore became the first direct spatial outputs produced by the implemented pipeline. Grounding DINO was selected on the basis of published benchmark performance rather than through an internal bench-

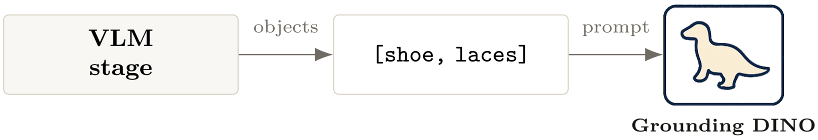


**Figure 3:** Simplified prompt handoff used by the implemented pipeline. The semantic stage emits a compact object list, and that list becomes the text prompt for Grounding DINO.

mark as extensive as the semantic VLM bakeoff. At the time of selection it represented the state of the art on zero-shot open-set detection, with published results reporting 52.5 AP on COCO without any in distribution training images and a mean 26.1 AP on the ODinW zero-shot benchmark, outperforming prior alternatives including GLIP and OWL-ViT [12]. In use, the stage showed the expected weaknesses of zero-shot grounding: small or visually similar objects could be confused, and category labels were not always assigned consistently. Nevertheless, the detector generally placed boxes in the correct image regions, which was the main requirement at this stage because those regions were then passed forward to mask generation. The archived run used the Hugging Face Grounding DINO post-processing default threshold of 0.25 for both box and text filtering, while the stricter 0.35 box threshold reported below follows the official Grounding DINO example configuration as a post-hoc sensitivity check [13, 14].

### 2.3 Object Mask Generation

The final implemented stage of the placement used SAM 2 after the bounding box step so that the detected objects also yielded binary masks [15]. This gave the pipeline denser spatial outputs than boxes alone and made the results more useful for later conditioning and inspection. The purpose of this stage was not to recover geometry or stable 3D identity. Its role was to provide a practical 2D segmentation signal aligned to the same windows already used by the semantic and grounding stages.

This distinction is central to how the placement should be interpreted. The mask stage belongs to an object pose estimation path only in a limited sense. What was implemented here was the bounding box and binary mask portion of that path, not the final geometry aware stage. That narrower reading is methodologically important because it explains both the value of the delivered work and the reason the 6 DoF stage was not forced into the placement before it was justified by the results.

In practical terms, integrating SAM 2 required implementing three concrete components: a box expansion step that enlarged detector outputs by a configurable pixel margin before passing them to the segmenter, a frame level batching loop that applied SAM 2 to each observation frame in the window, and a run length encoding serialiser that stored the resulting masks compactly together with their spatial shape metadata. Just as importantly, the mask rows remained aligned with the original semantic window order, which preserved a stable correspondence between captions, tags, detections, and masks for downstream reuse. SAM 2, which extends the original Segment Anything Model [16] to video and streaming contexts, was chosen largely from the literature and from its strong practical reputation as a general purpose segmentation model, rather than from a dedicated in house benchmark across many maskers.

This stage also clarified the practical interpretation of the outputs. A named limitation of the grounding and mask stages is category confusion between visually similar objects: qualitative inspection of notebook outputs showed cup-versus-lid misclassification in frames where both objects were present and similarly sized, consistent with the known sensitivity of zero-shot grounding to fine grained visual distinctions. In this placement context, that ambiguity did not invalidate the stage, because the value of the masks lay in providing spatially reasonable support regions for later conditioning rather than exact class level identification. This limitation becomes consequential only when the downstream objective requires precise object identity rather than approximate spatial localisation.

## 3 Results

### 3.1 Semantic Benchmark Results

The semantic benchmark expanded the model comparison first introduced in the one page progress report. Six candidate generators were evaluated: Qwen3-VL-2B, Qwen3-VL-8B, LLaVA-v1.6-7B, Idefics3-8B, Qwen3.5-4B, and SmolVLM2-2.2B. The evaluation sample consisted of 64 windows drawn deterministically from the EgoDex index using a fixed random seed, stratified to include a range of object categories and task phases. The benchmark slice was held fixed across all runs, with only the embedding backend varied, so that the contribution of the generator could be separated cleanly from the contribution of the similarity model. The sample size was appropriate for ranking candidate configurations rather than estimating absolute dataset-level performance; the OpenCLIP cross-family check was included to test whether the relative ordering remained stable under a second embedding space.

Table 1 summarises the main outcome across all evaluated generators. Both SigLIP and OpenCLIP scores are reported for each model; the OpenCLIP column is included as a cross family stability check rather than as a competing quality metric.

**Table 1:** Semantic benchmark results over the shared 64-window sample. SigLIP score and OpenCLIP score are cosine similarity proxies computed with each embedding family independently; they are not directly comparable across families. Lower seconds per window indicates lower runtime cost.

| Model | SigLIP score | OpenCLIP score | Sec./win |
|---|---|---|---|
| Qwen3-VL-2B | 0.365 | 0.409 | 1.432 |
| Qwen3-VL-8B | 0.431 | 0.473 | 2.435 |
| Qwen3.5-4B | 0.434 | 0.478 | 5.763 |
| LLaVA-v1.6-7B | 0.422 | 0.463 | 3.317 |
| Idefics3-8B | 0.405 | 0.450 | 4.186 |
| SmolVLM2-2.2B | 0.319 | 0.361 | 7.046 |

The comparison led to two practical results. First, SigLIP remained the benchmark-backed operational choice. The one page checkpoint and the final implementation both used SigLIP as the main embedding backend because its pairwise scoring behaviour is better suited to judging individual image–text pairs. OpenCLIP was retained only as a comparative check. Its scores were consistently higher than the matching SigLIP scores, but that does not mean the generated captions were better: OpenCLIP and SigLIP use different contrastive objectives and produce values on different scales [5, 4]. The OpenCLIP column therefore shows that the model ranking was broadly stable under a second embedding family, not that the pipeline should switch away from SigLIP.

Second, the Qwen family remained the strongest practical choice once quality and latency were considered together. Although larger models could improve raw semantic scores, they did so at a much higher runtime cost. For the placement, the most useful production choice remained Qwen3-VL-2B-Instruct together with SigLIP because it produced strong semantic outputs while keeping inference practical for repeated runs and later downstream stages. Figure 4 shows the quality versus runtime trade off that supported the final operational choice.

## 3.2 Grounding and Mask Results

The grounding and mask stages produced the first explicit spatial outputs from the placement pipeline. Using the semantic object list as the prompt source, Grounding DINO generated text conditioned bounding boxes, and SAM 2 then produced the corresponding binary masks. These results confirmed that the semantic stage was not only interpretable in isolation, but also useful as a prompt generator for downstream perception modules. Figure 5 shows an example output, while Figure 6 shows a second qualitative case with a different object configuration.

The strongest positive result here was that the semantic-to-grounding pipeline could be automated without freezing on manual prompt entry. To quantify this behaviour without rerunning inference or creating new annotations, the archived Grounding DINO and SAM 2 outputs were evaluated over a reconstructed 64-window proxy sample drawn from the full archived train part 1 run using the same fixed seed and a proportional task-stratified procedure. The exact original semantic benchmark window IDs were not recovered from the archive, so these results should be read as a conservative proxy evaluation rather than as the original benchmark slice.

**Table 2:** Grounding and mask proxy results over a reconstructed 64-window task-stratified sample. The 0.25 row reports the archive-native detector threshold, while the 0.35 row applies a stricter post-hoc score filter. These metrics measure whether the automated prompts produced usable spatial outputs; they are not precision, recall, or IoU against ground truth annotations.

| Metric | $\geq$0.25 | $\geq$0.35 |
|---|---|---|
| Window detection rate | 100.0% | 92.2% |
| Frame detection rate | 98.4% | 89.8% |
| Mean top-1 box score | 0.528 ± 0.139 | 0.544 ± 0.130 |
| Mask window rate | 100.0% | 92.2% |
| Mean SAM 2 mask score | 0.911 ± 0.089 | 0.923 ± 0.076 |

At the archive-native threshold, every sampled window produced at least one bounding box and one corresponding mask, with detections present in 126 of 128 sampled frames. Under the stricter 0.35 box-score filter, 59 of 64 windows still retained at least one detection and mask, and the five zero-detection cases were concentrated in add/remove-lid, airpod charging, tableware cleaning, and egg frying windows. The score distribution also showed that the result was not driven only by borderline boxes: 226 boxes remained above 0.35, including 78 above 0.50. The mask

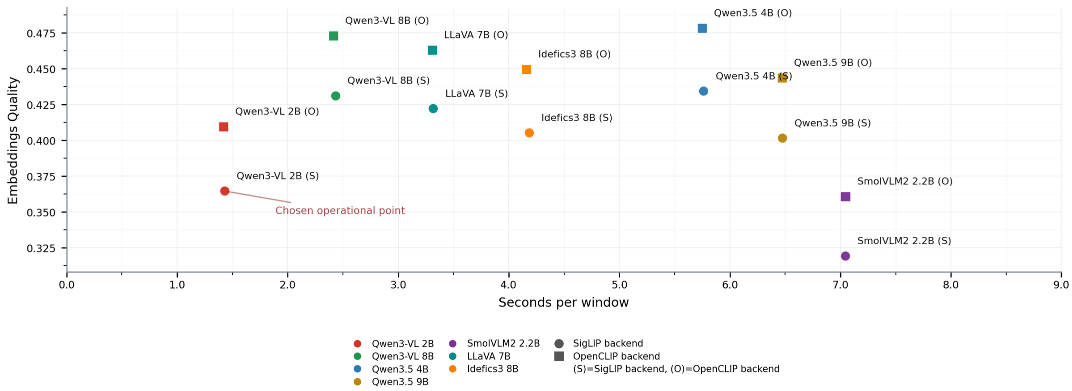


**Figure 4:** Semantic quality versus runtime trade off used to support the final production choice.

proxy was similarly usable, with a mean SAM 2 mask score of 0.923 at the stricter threshold and a mean mask-area fraction of 0.043 of the frame. A simple category-confusion proxy found five windows with overlapping boxes assigned to different prompt labels at IoU greater than 0.5, which is consistent with the qualitative observation that visually similar or co-located objects remain the main failure mode.

No formal precision or recall metric was computed against ground truth annotations, which remains a limitation of the current evaluation. Nevertheless, the proxy results support the central operational claim: semantic object lists are a practically viable automated prompt source, giving the placement a coherent multi stage workflow rather than a set of disconnected experiments.

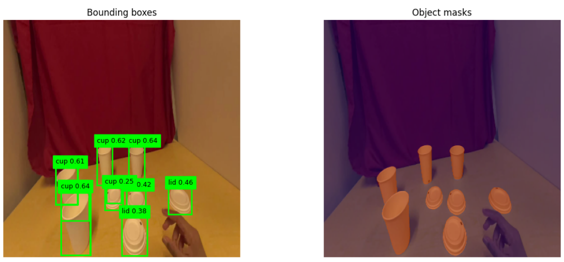


**Figure 5:** Example placement output from the grounding and mask stages. Semantic object prompts become bounding boxes and binary masks for the sampled window frames.

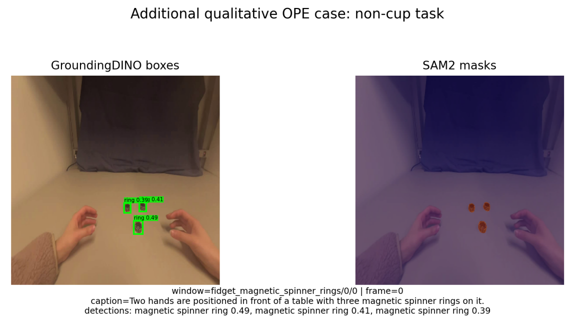


**Figure 6:** Additional qualitative Object Pose Estimation (OPE) pipeline example from a ring handling episode, included to show a second object configuration beyond the cup dominated case.

## 3.3 Limitations

Four principal limitations bound the scope of these results. First, the semantic benchmark was a bounded comparison over a 64-window shared sample; it is not a universal ranking and may not generalise to object categories or task types not represented in the sample. Second, the grounding and mask stages were evaluated using archived proxy metrics and qualitative inspection rather than against a formal annotated validation set with ground truth boxes or masks; no precision, recall, or IoU figures are reported. Third, the placement pipeline was developed and tested exclusively on EgoDex; cross dataset generalisation to other egocentric manipulation datasets was not evaluated. Fourth, although the pipeline produced enrichment data intended to reduce nuisance variation in downstream latent state learning, the placement did not yet train or evaluate a world model using those enriched signals. The report can therefore claim that the enrichment layer was produced and aligned, but not that it has already solved the noisy latent representation problem introduced in Section 1.

### 3.4 Interpretation

This bounded reading also explains the 6 DoF decision. By the time the semantic, bounding box, and mask stages had matured into a coherent prototype, the next step towards canonical 6 DoF pose recovery would have required a substantially different level of geometric machinery, stronger CAD dependence, and a more specialised evaluation setup [17, 18]. The placement therefore reached its best value by consolidating the 2D perception and conditioning pipeline rather than pushing prematurely into a heavier geometric stage.

## 4 Placement Effort and Supervision

The placement unit corresponds to approximately 100 to 140 hours of work. The hours worked over the placement period can be summarised as shown in Table 3, which presents a reconstructed allocation of 130 hours across the completed stages of the implementation. The allocation is estimated rather than logged: a conservative basis of 5 hours per day across 26 placement equivalent days was used, cross referenced against calendar records and implementation commit history. These activities were not always sequential, since model runtime, reruns, and waiting periods were interleaved with implementation, inspection, and refinement across multiple stages of the pipeline.

**Table 3:** Retrospective placement effort allocation reconstructed from the completed implementation stages.

| Placement activity | Days | Hours |
|---|---|---|
| Environment preparation, Docker checks, and `kr2280` setup | 1 | 5 |
| Semantic pipeline implementation and early validation | 6 | 30 |
| Object grounding implementation and prompt construction | 5 | 25 |
| Mask generation integration and refinement | 6 | 30 |
| Model rebenchmarking and later candidate checks | 4 | 20 |
| Notebook walkthroughs, result consolidation, and visual inspection | 2 | 10 |
| Report preparation and final packaging | 2 | 10 |
| Total | 26 | 130 |

This placement was also shaped by continuous supervision. The implementation state, benchmark outcomes, and main design decisions were communicated regularly to Prof. Alexiei Dingli throughout the placement period. This supervisory loop proved particularly important when new candidate models became available and when benchmark deltas required careful interpretation before any change to the operational pipeline was warranted.

## 5 Conclusion and Future Work

The strongest evidence for the value of this placement comes from its alignment with current research practice. As work within M.AI.ESTRO progressed and an academic paper currently in preparation undertook a systematic comparison of state-of-the-art world models, it became clear that the enrichment strategy central to this placement recurs independently across leading architectures. Within the M.AI.ESTRO programme itself, DexWM [19] encodes scene states using a frozen DINOv2 encoder whose semantically rich patch-level features serve the same function as the structured semantic outputs produced in this placement: both provide the enriched representation layer that a downstream world model predictor consumes. More broadly, PointWorld [20] enriches raw scene point clouds with frozen DINOv3 ViT-L visual features before world model conditioning, confirming that the same principle has been independently adopted at the frontier of open world robotic manipulation research. Together these two examples demonstrate that the enrichment strategy developed here is not an incidental design choice but a practice convergently validated by leading work in the field.

This placement successfully moved from raw EgoDex windows to structured semantic outputs, image space localisation, and binary masks, while benchmarking the model choices required to make that pipeline practical and reproducible. The contribution lies not only in the implemented stages but in the progression from a tightly coupled prototype to a modular, benchmark driven pipeline that can be extended or replaced at the component level. The work was carried out as supervised and iteratively reviewed development within the agreed project framework. Returning to the central research question posed in the Introduction, the evidence confirms that vision language model outputs can serve as reliable automated prompt sources for downstream open-set object grounding in egocentric manipulation data: the semantic object lists generated by Qwen3-VL-2B-Instruct triggered consistent detections across the reconstructed proxy sample without any manual prompt entry, satisfying the core condition the question set out to test.

The principal limitations of the delivered work are: the absence of a ground truth annotated validation set for formal grounding and mask evaluation; the sensitivity of zero-shot grounding to visually similar object categories; the restriction of all testing to EgoDex with no cross dataset evaluation; the absence of a downstream world model experiment confirming that the enrichment signals reduce nuisance latent variation; and the fixed stride windowing approach, which has since been superseded by an adaptive strategy in the broader research programme. These limitations define the boundaries of the current contribution and motivate the future work below.

Three immediate directions for future work follow from this placement. First, the semantic and open-set perception stages should be framed explicitly as a world-model enrichment layer. Rather than treating captions, object lists, boxes, and masks as isolated artefacts, the next implementation should expose them as a structured extraction layer that can be consumed by a downstream predictor. Second, that extraction layer can be used to build and benchmark an enriched world model variant in the same spirit as DexWM and PointWorld: the contribution would not be a direct copy of either architecture, but a M.AI.ESTRO-specific variant that tests whether semantic prompts, spatial detections, and masks improve predictive representations for industrial skill transfer. Third, the same methodology and reusable code can support the Malta–Poland Digital Product Passport research call, where structured perception and industrial AI methods are relevant to the Ecodesign for Sustainable Products Regulation [21]. Supporting work should also continue in parallel: a formally annotated subset of EgoDex windows would enable precision and recall evaluation of the grounding and mask stages; adaptive windowing should replace the initial fixed stride extraction end-to-end; and cross dataset evaluation on non EgoDex egocentric manipulation data would establish the generalisability of the semantic and grounding stages beyond the current source material. The placement thereby delivered a reusable, validated perception layer that connects raw egocentric demonstration data to the structured representations required by downstream learning and policy systems, establishing a foundation on which both the immediate M.AI.ESTRO objectives and the longer term research programme can build.

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